Ethical & Professional conduct

ICT 3209

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1. Ethics, Law and Professional Responsibility

Ethics: It refers to moral principles guiding right and wrong behavior.

Law: is a set of legally enforceable rules created by governments.

Professional Responsibility: It is the duty of professionals to act competently, honestly, ethically.

Example from computing:

- Ethics: A developer refuses to sell ~~dat~~ user data even if legal.
- Law: GDPR requires companies to protect user privacy.
- Professional Responsibility: A software ~~Engie~~ Engineer must write secure and reliable code.

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2. Histrical Ethical Failures in Technology.

Case I: Facebook - Combridge Analytica (2018)

User data was misused without consent for political targeting.

Case II: Therac-25 Radiation Machine

Software bugs caused deadly radiation overdoses.

Lesson: Safety-critical software must be rigorously tested and verified.

3. Software Engineering Ethics and Professional Responsibility.

Key principles:

- Public interest
- Quality software
- Honesty & Integrity
- Privacy & Protection
- Professional competence

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The ACM / IEEE code of Ethics guides engineer to prioritize, avoid harm, respect privacy, and be accountable in decision-making.

4. Utilitarianism vs Deontogy in computing

<u>Utilitarianism</u>: Focuses on greatest good for the greatest number.

<u>Deontology (Kantian Ethics)</u>: Focuses on duty or moral rules

<u>Privacy Dilemma</u>:

- Utilitarian: Allow data use if it benefits society.

- Deontolog: Reject unauthorized data use because it violets privacy right.

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5. Virtue Ethics in Software Engineering

Virtue Ethics focuses on character traits like honesty, fairness and responsibility.

- A virtuous engineer avoids unethical shortcuts. ensure data security and acts with integrity even under ~~ge~~ pressure.

6. Explainability in High-stocks AI system:

Explainability ensures AI decisions are understandable and transparent

If AI is black box:

- Harm1: Discrimination in loan approv
- Harm1: Wrongful criminal sentencing due to biased algorithm

7. ACM code of Ethics & it's importance:

Key principles:

I. Contribute to society

II. Avoid harm

III. Be honest & trustworthy

IV. Respect Privacy

V. Maintain competence

8. Whistleblowing in computing: conflict between social good and privacy

Scenario: A data scientist uses health data to predict healt data.

conflict: Public health benefit vs Personal ~~pri~~ privacy

ACM guidance: Minimize data use, anonymize data ~~do~~ use and obtain consent.

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9. Whistleblowing in computing

Whistleblowing is reporting unethical or illegal activities within an organization

Challenges : job loss, legal risk, workplace pressure

Ethical duty : Protect public interest & prevent harm.

10. Intellectual property Rights in software

- Copyright : Protects codes & software contents
- Patent : protects new algorithms or inventions
- Trademark : Protects brand name and logos.

11. Open Source vs proprietary licensing

<u>open source</u>: Free to use, modify and distribute (linux)

<u>proprietary</u>: Restricted and owned by a compony (e.g., windows)

12. Professional Dissent and whistleblowing

<u>Professional Dissent</u>: Internal disagreement about unethical pratices

<u>whistleblowing</u>: public disclosure of wrongdoing

13. Informed consent and Data collection

Informed consent means users clearly understand how their data is collected and used

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14. Ethics of surveilance Technology

Concerns:

(i) privacy invasion

(ii) Government misuse

(iii) Mass monitoring

Example: Facial recognition used for unauthorized tracking

15. Copyright vs Patent in Technology computing

copyright: protects software code & UI design

patent: protects innovative algorithms or technical inventions.

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16. Algorithmic bias & Fairness

Algorithmic bias occurs when AI produces unfair outcomes due to biased data or design

causes:

(I) Biased datasets

(II) Poor model design

Solutions

(I) Diverse datasets
(II) Fairness testing
(III) Regular audits

17. Accountability in Autonomous system

Transparency and explainability are essential for trust and safety

Accountability should be shared among

- Developer
- companies
- Regulators